Homework 08

AUTHOR
cgao19

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All work is in Python. Both questions use the supplied score functions as given; nothing is fitted.

▶ Show the workflow

Question 1: Boundary and margin

Part (a): distances and functional margins

The score is $f(x) = \boldsymbol{\beta}^{\mathrm{T}} x$ with $\boldsymbol{\beta} = (0.8, 0.6)^{\mathrm{T}}$. The signed distance from x to the hyperplane $f(x) = 0$ is $f(x) / \| \boldsymbol{\beta} \|_2$, and here $\| \boldsymbol{\beta} \|_2 = \sqrt{0.64 + 0.36} = 1$, so the unsigned distance is simply $|f(x)|$ and the functional margin $yf(x)$ *is* the signed distance measured toward the observation's own class. That coincidence is what makes the lines $f(x) = \pm 1$ sit exactly one unit away. The three cases follow from comparing $yf(x)$ with 1:

$$y_i f(x_i) < 1 \;\Rightarrow\; \text{inside}, \qquad y_i f(x_i) = 1 \;\Rightarrow\; \text{on}, \qquad y_i f(x_i) > 1 \;\Rightarrow\; \text{beyond}.$$

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```
||beta|| = 1.0, so distance = |f(x)| and y f(x) is the signed distance
```

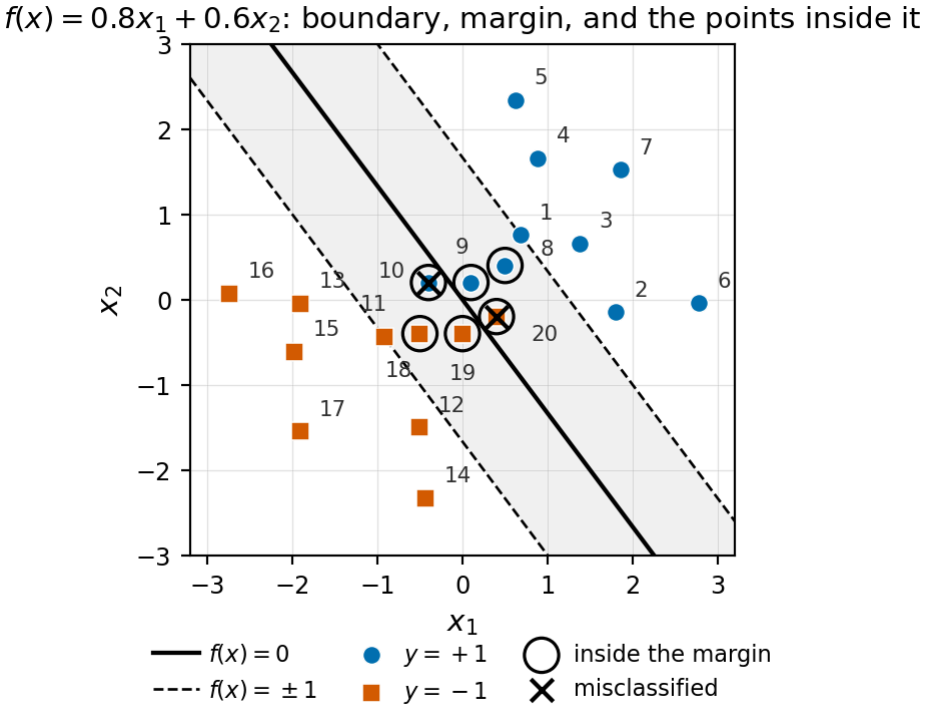
	x1	x2	y	f(x)	distance	y f(x)	position	side
obs								
1	0.68	0.76	1	1.00	1.00	1.00	on	correct
2	1.80	-0.15	1	1.35	1.35	1.35	beyond	correct
3	1.38	0.66	1	1.50	1.50	1.50	beyond	correct
4	0.88	1.66	1	1.70	1.70	1.70	beyond	correct
5	0.62	2.34	1	1.90	1.90	1.90	beyond	correct
6	2.78	-0.04	1	2.20	2.20	2.20	beyond	correct
7	1.86	1.52	1	2.40	2.40	2.40	beyond	correct
8	0.50	0.40	1	0.64	0.64	0.64	inside	correct
9	0.10	0.20	1	0.20	0.20	0.20	inside	correct
10	-0.40	0.20	1	-0.20	0.20	-0.20	inside	wrong
11	-0.92	-0.44	-1	-1.00	1.00	1.00	on	correct
12	-0.50	-1.50	-1	-1.30	1.30	1.30	beyond	correct
13	-1.90	-0.05	-1	-1.55	1.55	1.55	beyond	correct
14	-0.44	-2.33	-1	-1.75	1.75	1.75	beyond	correct
15	-1.98	-0.61	-1	-1.95	1.95	1.95	beyond	correct
16	-2.74	0.07	-1	-2.15	2.15	2.15	beyond	correct
17	-1.90	-1.55	-1	-2.45	2.45	2.45	beyond	correct
18	-0.50	-0.40	-1	-0.64	0.64	0.64	inside	correct
19	0.00	-0.40	-1	-0.24	0.24	0.24	inside	correct
20	0.40	-0.20	-1	0.20	0.20	-0.20	inside	wrong

Observations 1 and 11 land on the margin lines exactly ($yf(x) = 1$), the data having been constructed that way. Observations 10 and 20 have *negative* functional margins, -0.20: they are 0.20 from the boundary but on the far side of it, so they are both inside the margin and misclassified.

Part (b): the boundary and margin band

Writing $f(x) = 0.8x_1 + 0.6x_2$, the three lines $f(x) \in \{-1, 0, 1\}$ are $x_2 = (c - 0.8x_1)/0.6$ for $c = -1, 0, 1$, which is what the plot draws.

▶ Show the workflow



The 20 observations with the boundary f(x) = 0 (solid) and the margin lines f(x) = +/-1 (dashed). Colour gives the true class; black rings mark the six observations inside their margin, and crosses mark the two that the rule misclassifies. The shaded band is the margin. Axes are equal in aspect, so the perpendicular gap between the boundary and either dashed line really is 1 unit.

The picture matches the table. The band between the dashed lines is empty except for the six circled points, and the two crossed points sit across the solid line from their own colour.

Part (c): counts and misclassifications

▶ Show the workflow

```
misclassified (y f(x) < 0): 10, 20
in-sample error rate: 0.10 = 2/20
slacks of the margin violators: {8: 0.36, 9: 0.8, 10: 1.2, 18: 0.36, 19: 0.76, 20: 1.2}
```

	count	observations
position		
inside	6	8, 9, 10, 18, 19, 20
on	2	1, 11
beyond	12	2, 3, 4, 5, 6, 7, 12, 13, 14, 15, 16, 17

Six observations are inside the margin (8, 9, 10, 18, 19, 20), two are exactly on it (1 and 11), and twelve are beyond it. Only observations **10** and **20** are misclassified, an error rate of $2/20 = 10\%$. Both are also inside the margin, which is the general rule: a misclassified point has $yf(x) < 0 < 1$, so every misclassification is necessarily a margin violation, while the converse fails — observations 8, 9, 18, and 19 are correctly classified yet still intrude into the band. In soft-margin SVM language, all six circled points would carry a positive slack $\xi_i = 1 - y_i f(x_i)$, ranging from 0.36 for observations 8 and 18 up to 1.20 for observations 10 and 20, and only the latter two exceed $\xi_i = 1$.

Question 2: Margin in four dimensions

Part (a): projection onto β and the geometric margin

Every point of a hyperplane's score depends on $\boldsymbol{x}$ only through its component along $\boldsymbol{\beta}$, so the whole four-dimensional configuration collapses losslessly onto the single coordinate

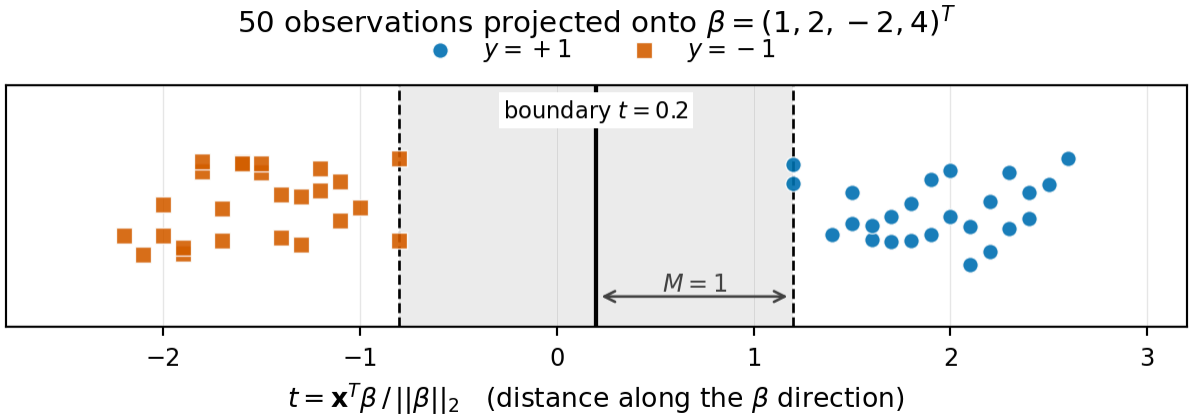
$$t_i = \frac{\boldsymbol{x}_i^{\mathsf{T}}\boldsymbol{\beta}}{\| \boldsymbol{\beta} \|_2}, \qquad \| \boldsymbol{\beta} \|_2 = \sqrt{1+4+4+16} = 5,$$

on which the boundary $\beta_0 + \boldsymbol{x}^{\mathsf{T}}\boldsymbol{\beta} = 0$ is the single point $t = -\beta_0 / \| \boldsymbol{\beta} \|_2 = 1/5 = 0.2$. Signed distance to the boundary is then $t_i - 0.2 = f(\boldsymbol{x}_i)/5$, and the geometric margin is $M = \min_i y_i f(\boldsymbol{x}_i) / \| \boldsymbol{\beta} \|_2$.

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```
||beta|| = 5   boundary at t = -beta0/||beta|| = 0.2
projection spans t in [-2.2, 2.6]; misclassified: 0
smallest functional margin  min y f(x) = 5  (observations [4, 15, 33, 36])
geometric margin  M = 5 / 5 = 1.0
margin band on the t axis: [-0.8, 1.2], width 2M = 2.0
```

▶ Show the workflow



The 50 observations projected onto the direction of beta. The horizontal axis is t = x'beta/||beta||, a true distance axis; vertical jitter only separates the markers. The solid line is the boundary at t = 0.2 and the dashed lines are the closest points of each class, at t = -0.8 and t = 1.2. The classes are perfectly separated and the nearest point on either side is exactly 1 unit away, so M = 1.

▶ Show the workflow

the six observations closest to the boundary

	y	t	f(x)	y f(x)	distance
obs					
17	-1	-1.00	-6.00	6.00	1.20
15	-1	-0.80	-5.00	5.00	1.00
36	-1	-0.80	-5.00	5.00	1.00
4	1	1.20	5.00	5.00	1.00
33	1	1.20	5.00	5.00	1.00
31	1	1.40	6.00	6.00	1.20

Reading the margin off the plot: the rightmost $y = -1$ point sits at $t = -0.8$ and the leftmost $y = +1$ point at $t = 1.2$, each exactly $0.2 - (-0.8) = 1.2 - 0.2 = 1$ away from the boundary, so

$$M = \min_i \frac{y_i f(\boldsymbol{x}_i)}{\| \boldsymbol{\beta} \|_2} = \frac{5}{5} = 1.$$

Four observations attain it — 15 and 36 from the negative class, 4 and 33 from the positive class — and these are the points that would be support vectors if this boundary were the maximum-margin one. The two classes are perfectly separated on the projection (no point has $y_i f(\boldsymbol{x}_i) < 0$) and the empty band is symmetric about the boundary, so this given hyperplane is already sensibly placed, not merely separating.

Part (b): the SVM normalization

The pair $(\boldsymbol{\beta}, \beta_0)$ and any positive rescaling $(c\boldsymbol{\beta}, c\beta_0)$ describe the same set $\{f = 0\}$, so the hyperplane is identified only up to scale. SVM fixes the scale by demanding $\min_i y_i f(\boldsymbol{x}_i) = 1$. Since the current minimum is $\min_i y_i f(\boldsymbol{x}_i) = 5$, dividing through by 5 does it:

$$\tilde{\boldsymbol{\beta}} = \frac{\boldsymbol{\beta}}{5}, \qquad \tilde{\beta}_0 = \frac{\beta_0}{5}.$$

▶ Show the workflow

```
scaling constant c = min y f(x) = 5
beta tilde  = (0.2, 0.4, -0.4, 0.8)'
beta0 tilde = -0.2
||beta tilde|| = 1.0000   1/||beta tilde|| = 1.0000   (equals M = 1.0000)
new smallest functional margin: 1.0000
labels unchanged: True; distances unchanged: True
```

So the same decision boundary in SVM normalization is

$$\tilde{\boldsymbol{\beta}} = (0.2,\ 0.4,\ -0.4,\ 0.8)^{\mathsf{T}}, \qquad \tilde{\beta}_0 = -0.2,$$

that is $\tilde{f}(\boldsymbol{x}) = -0.2 + 0.2x_1 + 0.4x_2 - 0.4x_3 + 0.8x_4$. Two checks confirm it is the same hyperplane seen at a different scale: every predicted label is unchanged, and every geometric distance is unchanged, because both the numerator and the denominator of $y_i f / \| \boldsymbol{\beta} \|$ were divided by 5. What the normalization buys is the identity $\| \tilde{\boldsymbol{\beta}} \|_2 = 1/M$: here $\| \tilde{\boldsymbol{\beta}} \|_2 = 1$ exactly, matching $M = 1$ from part (a). This is the reason the SVM optimization can be stated as minimizing $\| \boldsymbol{\beta} \|_2^2$ subject to $y_i f(\boldsymbol{x}_i) \geq 1$ — under this normalization shrinking $\| \boldsymbol{\beta} \|_2$ *is* widening the margin.

Question 3: Peer review of a Homework 7 skill

Part 1: using the assigned skill

The assigned student's Homework 7 report contains the skill under the heading *LDA/QDA Analysis and One-Page Report*, with the sections **Behavior**, **Analysis**, and **Report requirements**. Two things complicated copying it "without changes":

- the listing is **rendered Markdown** (headings, bullets, emphasis) rather than a verbatim code block, so some source characters are unrecoverable — the blend formula arrives as *p = w p_LDA + (1-w) p_QDA* because the Markdown asterisks were consumed as emphasis;
- **no YAML frontmatter appears anywhere in the report**; `name:` and `description:` are absent. A `SKILL.md` without frontmatter cannot register as a skill, so it can only be supplied to the agent as a plain instruction file.

I reconstructed the file as faithfully as the rendered output allows and saved it to `peer-skill/SKILL.md` (4,674 bytes). The reconstruction loses Markdown markup, so the author's original is somewhat larger; the report states the size only as "5Kb", against a limit of 5,000 bytes.

The skill was supplied by path, together with the dataset, the response, and the predictor coding, and with no preferences of my own about methods, figures, model selection, or report style.

Run log. Executed 2026-10-18, 17:20:06 to 17:29:48 — 9 minutes 42 seconds of wall-clock time, of which the statistical work (twelve candidates, 25 repeated-CV fold estimates each, and a 25-fold outer nested CV) took 57 seconds. The remaining eight minutes went to building the one-page PDF the skill demands and to two rounds of layout repair. Token use stayed modest and at no point approached the weekly quota, so **no steer message was sent and the run was not stopped early**. Artefacts are in `peer-skill-run/`.

The skill drove the analysis to a defensible result without any methodological input from me. It selected PCA to six components followed by LDA, with a nested-CV prediction error of **0.2649, 95% interval [0.2511, 0.2787]**, against a majority-class baseline of 0.3463, and it produced the one-page PDF with the two-panel figure, the twelve-row table with log-loss column, the Findings paragraph and the Limitations paragraph that its own Report requirements section specifies. Three frictions appeared while executing it:

- **It assumes more than one discriminant axis.** Figure panel (b) asks for "the first two discriminant coordinates" and step 6 for "separation share per discriminant axis". With $K = 2$ there is exactly one such axis, carrying 100% by construction. The skill's parenthetical "(or principal)" rescued the figure, but the separation-share item is vacuous for every two-class problem — the commonest case it will meet.

- **The one-standard-error tie-break decided the outcome, and it points the wrong way.** This is item 2 of the review below.
- **Two choices are left undefined:** the CV scheme for the *inner* loop of the nested CV, and where the report should be written. Both fell to me, which is exactly the kind of discretion the skill otherwise works hard to remove.

The skill's overflow rule — "shorten prose first, then shrink the figure; never drop the figure, table, or numerical results" — proved genuinely useful: the first draft ran to two pages and the rule said unambiguously what to cut. Its instruction to render the PDF to an image and check it also earned its place, catching a clipped axis annotation and a wrapped table column that the page-count check alone would have missed.

Part 2: peer review

Peer review of the Homework 7 LDA/QDA skill

Overall assessment. The skill is thoughtfully designed: it keeps the model family correct, fits preprocessing inside CV folds, applies a one-standard-error rule, and asks for a nested-CV error estimate because selection biases the plain CV number. Running it myself with no added preferences, it worked — nested-CV error 0.2649, 95% interval [0.2511, 0.2787], and every element of the required one-page report. The problems below are narrower: the analysis attached to Homework 7 skipped most of what the skill demands, one tie-break rule backfires, and a couple of its conclusions overreach the data.

1. The skill requires deliverables that the delivered report does not contain. The skill asks for a two-panel figure, a log-loss column, a nested-CV interval, the majority-class baseline, and a Limitations paragraph. None of these appear in the Homework 7 report, which has no figure, a three-column table, and stops after Findings; it also used a single 20% hold-out rather than the nested CV the skill specifies. A skill's value is the guarantee it gives its user, and a silently dropped requirement gives none. I'd add a short pre-delivery checklist as the skill's final step — "confirm each of: figure, log-loss column, nested-CV interval, baseline, limitations, or state why one was skipped" — since agents follow a closing checklist more reliably than instructions buried mid-document.

2. The one-standard-error tie-break backfires: it picks the more complex model. When I ran the skill, two candidates tied exactly at a CV error of 0.2662 — plain shrunken LDA and a PCA-then-LDA pipeline — and the rule's preference for "fewer PCA components" chose the PCA pipeline, because a candidate with no PCA stage is apparently not credited with having fewer components. The nested loop repeated this choice in 23 of 25 outer folds, so it is systematic. This is the opposite of what a one-standard-error rule is for. I'd replace the three-key ordering with one explicit complexity measure, such as the number of fitted parameters.

3. The test-set conclusion is stated more confidently than the sample size allows. The report says LDA "performed best" because it misclassified 22 of 93 test observations against QDA's 24 — a two-case gap that is well under one standard error on either rate. The CV table also labels a per-fold SD as "SE", though the one-standard-error rule needs the smaller SE of the mean. The real, defensible finding is that LDA's error sits meaningfully below the 34.4% majority-class baseline; the LDA-versus-QDA comparison itself is not resolvable at this sample size. I'd report errors with intervals and lead with the baseline comparison rather than the LDA–QDA gap.

Part 3: the GitHub issue

The review above was posted as a GitHub Issue in the assigned student's Homework 7 repository: github.com/raindrop110/stat432-fall2026/issues/5.